

Experiment No. 20 (Mixture No. 1)

Date: / /

Aim : Analyse two acidic (anion) radicals qualitatively from given inorganic mixture.

Apparatus : Test tubes, test tube holder, test tube stand, filter paper etc.

A. Preliminary tests:

Test	Observation	Inference
Color	white / colourless	chloride, nitrogen sulphate, carbonates of Ca^{++} , Ba^{++} , Sr^{++} or Mg^{++} may be present.
Nature	Crystalline (Soluble in water)	Cl^- , Br^- , I^- , NO_3^- , SO_4^{--} , CO_3^{--} , CH_3COO^- or compounds of NH_4^+ or K^+ may be present.

B. Dry Tests for Acidic radicals (Test tube must be dry for following tests)

Test	Observations	Inference
1. Heating in a dry test tube:- Take a small quantity of the mixture in a clean and dry test tube and heat it strongly	colourless odourless gas which turns lime water milky	CO_3^{--} , $\text{CO}_3\text{O}_4^{--}$ may be present.
2. Action of dil. H_2SO_4 : Take a small quantity of mixture + dilute H_2SO_4	Evolution of colourless odourless gas with brisk effervescence of CO_2 gas which turns freshly prepared lime water milky	CO_3^{--} may be present
3. Action of conc. H_2SO_4 : Take a small quantity of mixture + conc. H_2SO_4	No evolution of any gas	Cl^- , Br^- , I^- , NO_3^- , $\text{CO}_3\text{O}_4^{--}$ are absent
4. Action of Cu foils and conc. H_2SO_4 Mixture + Cu filings and conc. H_2SO_4 , heat it strongly	—	—
5. Action of MnO_2 and conc. H_2SO_4 Mixture + MnO_2 powder + conc. H_2SO_4	—	—

Individual dry tests for SO_4^{--} & PO_4^{--} (if required)

1. Test for sulphate Mixture + dil. HCl , boil the solution + BaCl_2 solution	white ppt. insoluble in conc. HCl	SO_4^{--} may be present
2. Test for phosphate Mixture + conc. HNO_3 , boil + excess of ammonium molybdate solution	—	—

C. Preparation of original solution (O.S) : Mixture is dissolved in 20 mL quantity of distilled water in a beaker, stir with glass rod to dissolve the mixture. Clear solution is obtained. Use this O.S. for further tests of acidic radicals.

D. Wet Tests for Anion (Acidic Radicals)

1. O.S. + AgNO_3	white ppt. Soluble in dil HNO_3 with effervescence of CO_2	CO_3^{2-} may be Present
If white ppt obtained in above test, perform following distinction test		
Distinction between Cl^-, Br^- and I^- : O.S. + dil. H_2SO_4 (till acidic) + chloroform + Cl_2 water (fresh) in excess, shake vigorously and observe the colour of chloroform layer carefully.		
2. O.S. + $\text{Ba}(\text{NO}_3)_2$ solution	white ppt + insoluble in dil HNO_3	SO_4^{2-} Present
3. O.S. + dil. acetic acid + (Freshly prepared) FeSO_4 solution		
4. O.S. + diphenyl amine + conc. H_2SO_4		
5. O.S. + FeCl_3 solution		
6. O.S. + dil. acetic acid + CaCl_2		
7. O.S. + dil. H_2SO_4 + 2-3 drops of KMnO_4		

E. Confirmatory tests for Acidic Radicals

1. C.T. for first detected acidic radical CO_3^{2-} (carbonate)

Test	Observation	Inference
1. 0.5 + 2-3 drops of Phenolphthalein indicator	Pink colour	CO_3^{2-} confirmed
2. 0.5 + $\text{Ba}(\text{NO}_3)_2$	white ppt soluble in dil HNO_3	CO_3^{2-} confirmed

2. C.T. for second detected acidic radical

Test	Observation	Inference
1. 0.5 + lead acetate solution	white ppt. insoluble in dil HNO_3	SO_4^{2-} sol ⁿ confirmed
2. 0.5 + BaCl_2	white ppt of BaSO_4	SO_4^{2-} confirmed

Result :

The given inorganic mixture no. 1 contains following two anions (Acidic radicals)

i) CO_3^{2-} (Name of anion) (carbonate)

ii) SO_4^{2-} (Name of anion) (carbonate)

Remark and sign of teacher: